

Quantifying Latent Cognitive Processes of Encoding and Retrieval with an Updated Hierarchical Bayesian Cognitive Processing Model, Including Prior Data and Adjustments Tuned for the ADAS-Cog Word Recall Test

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Summary

Embic Corporation has previously provided quantified cognitive processes (**qCPs** – formerly expressed as Digital Cognitive Biomarkers or DCBs) for all ADAS-Cog Word Recall and Delayed Word Recall tests, collectively wordlist memory (**WLM**) tests, done under ADNI1, ADNI-GO, ADNI2, and ADNI3 (data file: ADNI_EMBICDCB.csv). This method document provides an overview of new qCP generation and summarizes the enhancements made to adapt our parameter modeling for features of the ADAS-Cog WLM test and its multiple wordlists (datafile: ADAS_EMBIC_qCP.csv).

Embic's qCPs quantify unobservable underlying (latent) cognitive processes of encoding and retrieval. The qCPs are generated with a hierarchical Bayesian cognitive processing (**HBCP**) model that analyzes item response data from commonly used WLM tests of learning and recall.

The original version of the HBCP (**HBCP-1**) generated, for the ADAS-Cog WLM test, seven base qCPs, each representing the probability of information processing through different encoding (N_1 , N_2 , N_3 , or N_4) or retrieval (R_1 , R_2 , or R_3) pathways, to and from three distinct storage states (*pre-task*, *transient*, or *durable* storage states).

While the HBCP-1 qCPs account for demographic characteristics and include adjustments for learning task item presentation position, they were not adjusted for the effect of three different wordlists used across ADNI1, ADNI-GO, ADNI2, and ADNI3. The new HBCP model (**HBCP-2**) quantifies the recall of each word independently from one another, an important feature that accounts for the shuffled presentation order used in the ADAS-Cog, in a way that produces latent cognitive process parameters specific to the participant at time of assessment and separate from test wordlist features. The HBCP-2 also incorporates normative prior data and item presentation position adjustments that are better tuned to the ADAS-Cog WLM test features.



Accompanying this document is a datafile, ADAS_EMBIC_qCP.csv, containing all qCPs and behavioral M parameters calculated for each ADAS-Cog WLM assessment for 2,366 participants (10,868 assessments) assessed under ADNI1, ADNI-GO, ADNI2, and ADNI3, dates ranging from 5/17/2005 to 04/07/2023.

Background

Hierarchical Bayesian Cognitive Processing Model

The hierarchical Bayesian cognitive processing (**HBCP**) model is applied to item response data from commonly used wordlist memory (**WLM**) tests of learning, recall, and recognition. Application of this model improves the use of information collected through WLM tests and enables characterization of more subtle but distinct cognitive changes at the unobservable processing level that are not quantifiable with observed behaviors directly [1].

Embic's quantified cognitive processes (**qCPs** – previously expressed as Digital Cognitive Biomarkers or DCBs) are generated with such an HBCP model and characterize the unobservable (latent) cognitive processes of encoding and retrieval. There are seven base qCPs, each representing the probability of information processing through different encoding (N_1 , N_2 , N_3 , or N_4) or retrieval (R_1 , R_2 , or R_3) pathways, to and from three distinct storage states (*pre-task*, *transient*, or *durable* storage states).

Figure 1 shows how items on the WLM test move among storage states via the encoding qCPs and how they are recalled from them via the retrieval qCPs, across the tasks of the test. When data from a delayed recognition task are available, the model incorporates it for additional precision. Normative prior information is also provided to the model from evaluation of a large dataset of participants, which were sampled from a database of nearly 2,000,000 individuals without diagnosis of cognitive impairment in the general population.

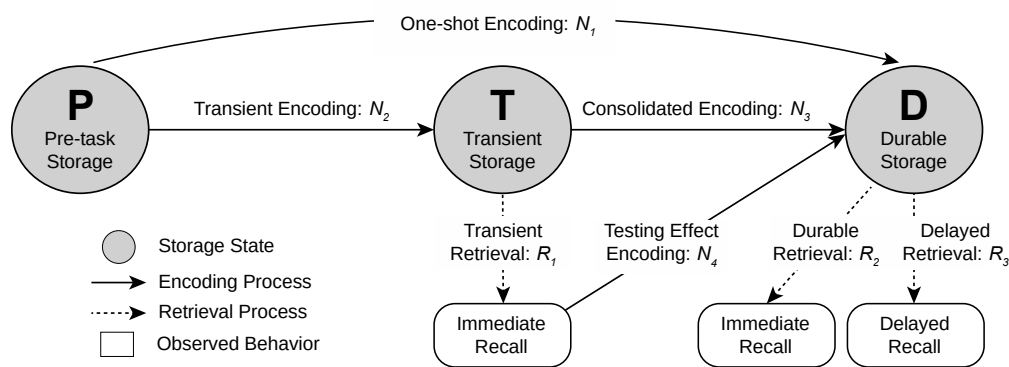


Figure 1. Hierarchical Bayesian Cognitive Processing Model. The model has three episodic memory storage states (P, T, and D), four processes of encoding into them (N_1 , N_2 , N_3 , and N_4), and three processes of retrieval from them (R_1 , R_2 , and R_3).

Importance of Considering Different Wordlist Features

When scoring WLM tests (e.g., ADAS-Cog Word Recall and Delayed Recall, AVLT, EVLT, and CERAD), it is essential to consider different features of those tests. Such features include, but are not limited to, the number of list words, the list presentation order, the presence of semantic categories, and the number of learning and recall tasks. They greatly and differently impact the way that individuals learn and remember the words presented, thus affecting their test performances. Adjusting the scoring for these differences enables characterization of latent ability for encoding and retrieval, independent of the test that those abilities were measured with. This allows these abilities to be compared across individuals and across measurements within the same individual, even when using different WLM tests. Further details on this subject are discussed elsewhere [2].

Wordlist Feature Differences Identified in ADNI ADAS-Cog WLM Tests

The ADAS-Cog Word Recall and Delayed Word Recall test, a WLM test used in ADNI, presents a 10-word list in a shuffled order across three learning tasks. A serial position curve (i.e., increased recall for list-beginning items through *primacy* and increased recall for list-ending items through *recency*), common to episodic list-learning tasks, can be observed in the recall of items on the first immediate recall task. However, because list word presentation position changes from task to task, serial position effects are obscured across the full test. As a result, the shuffling of the list words removes the ability to measure the influences of primacy and recency on an individual's behavior, and these influences have been found to be characteristic of changes due to Alzheimer's disease progression [3]. This shuffling also requires any model that uses position-specific item response data to adjust for these changing positions to describe learning across tasks.

Additionally, ADNI uses three different wordlists across ADNI1, ADNI-GO, ADNI2, and ADNI3. Our comparative analysis demonstrates that these wordlists are not equivalent, and, without adjustment for the influence of the word features within these lists, they impact both word recall ability and qCP generation within and between individuals assessed with different wordlists (**Figure 2**).

Distributions of Individual Assessment Mean Unadjusted qCP across Positions, by Wordlist

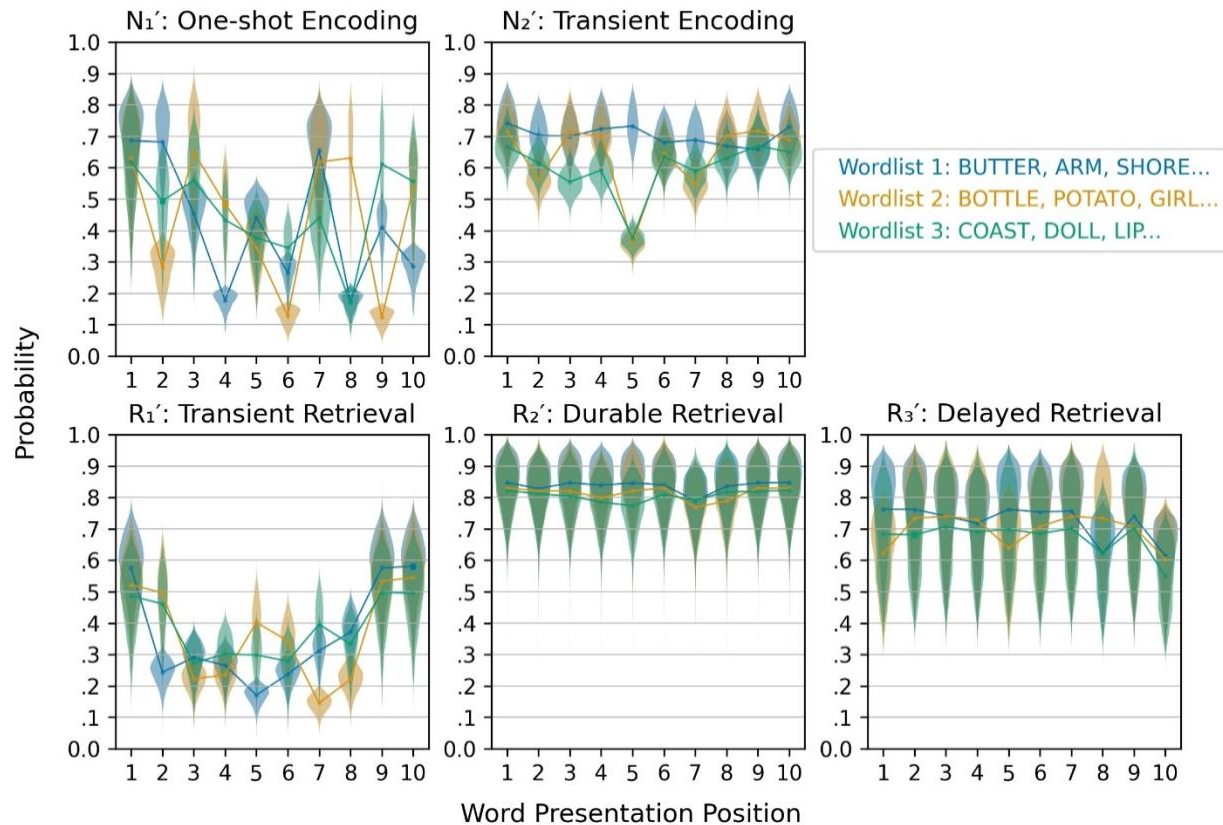


Figure 2. Probabilities of participants' encoding and retrieving words on the WLM test, including both latent ability and the relative difficulty of the items that fall within each position in the list presentation. Shuffling in the ADAS-Cog WLM results in erratic patterns (N_1), noisy serial position curves (R_1), or flat patterns where decline would be expected at the end of the list (R_3). While the lists are balanced when averaged across positions, they are non-equivalent when considering per-position item response data.

Since the presentation positions and word features of the three different wordlists used in ADNI ADAS-Cog WLM tests affect test performance, the HBCP-1, the model that was used to generate the original qCPs (data: ADNI_EMBICDCB.csv), was further updated to address adjustments required for position and wordlist effects as the HBCP-2 and generated new qCPs (data: ADAS_EMBICqCP2.csv).

HBCP-2 Model

To accommodate ADAS-Cog WLM tests' feature differences in ADNI, several updates were made to the HBCP-1 model.



Generating New Priors Accounting for Word Presentation Order

The techniques developed to account for word presentation order (position penalties) in the ADAS-Cog WLM test were applied to prior-generating models, using Embic's Verbal Learning Test (**EVL**T) data from a neurology clinic (N = 2,047) and insurance underwriting (N = 63,065) samples. Priors were generated in hierarchical models for each combination of sex (female and male), education (high school education or less and greater than high school education), and age (in 5-year intervals, from 45 to 85) groups, each with 2,000 participants, excepting two groups, 1) 45-to-50-year-old males with high school education or lower (n = 1,859) and 2) 80-to-85-year-old males with high school education or lower (n = 1,206). Though these data were from fixed-order tests (i.e., the presentation of the 10 words remained the same across learning tasks), an independent model could be and was applied without any assumptions or constraints for serial position effect, because such an independent model is required for shuffled-order presentation WLM tests. By using the same independent model throughout the process, more precise priors with a greater degree of difference across demographic groups were obtained. These priors inform the model based on a participant's age, sex, and education, and any non-missing combination of the three (**Figure 3**).

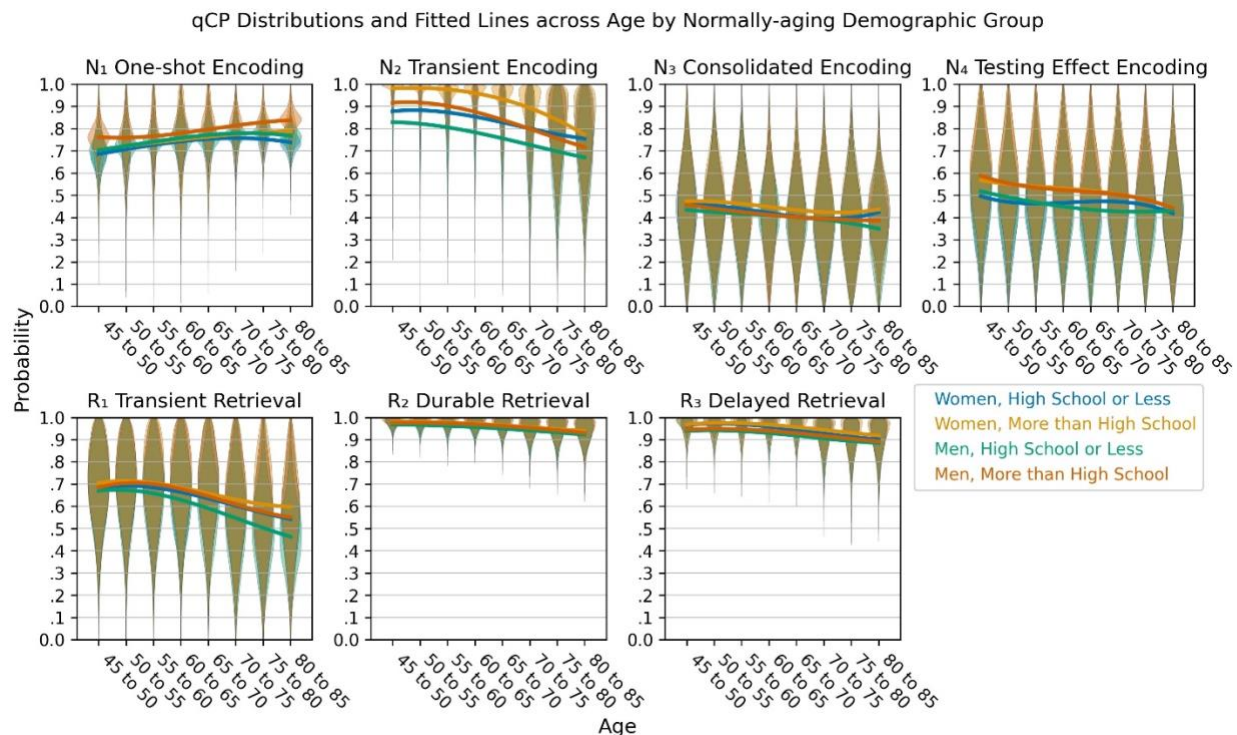


Figure 3. Normative data generated for each qCP and for each demographic group, divided across age, sex, and education level, to be used as prior data in HBCP-1 models. Differences across demographic groups are especially apparent in N_1 and N_2 .



Generating Position Penalties

Using these updated priors, we re-evaluated the position penalties for the ADAS-Cog wordlists and better characterized the impact of shuffling order and the differences across the three wordlists used in ADNI data. HBCP-2 models were applied to a subset of the ADNI data, using only assessments at the first exposure to each wordlist for each individual while cognitively normal (Wordlist 1 $n = 875$; Wordlist 2 $n = 466$; Wordlist 3 $n = 197$), and from them generating distributions of penalties to apply to each position for each qCP for each wordlist. Mean penalties were recorded and are used in subsequent modeling. These penalties are factored into the values presented in **Figure 2**.

Generating New qCPs

With both updated priors and updated position penalties, the seven base qCPs were generated on all ADNI assessments that had 1) full item response data to the ADAS-Cog WLM test and 2) one of the three specific wordlists identified in the data. These qCPs are described in the data delivery in terms of means and standard deviations for the individual assessment. The distributions these means and standard deviations generate represent the probable range of a participant's true qCP, and researchers using them are encouraged to account for this degree of uncertainty in analyses. This can be done by sampling from a truncated (between 0 and 1) normal distribution with the participant's mean and standard deviation or by performing comparative analyses with a lower bound estimate (e.g., $M - 1 SD$), the mean, and an upper bound estimate from all participants. Mean qCPs are nevertheless considered the best single value and can be used as-is.

Recall Behavior Parameters (M)

A set of M parameters was generated along with the new qCPs, which represent the expected recall rate of an individual as a behavioral measure. Six M s are included in the ADNI data delivery (data: ADAS_EMBICqCP2.csv), four of which characterize an individual's recall rate on a specific task of the ADAS-Cog (M_{IFR1} , M_{IFR2} , M_{IFR3} , and M_{DFR1}) and two of which characterize an individual's recall rate through retrieval from the *transient* or *durable* storage states in the HBCP model (M_{Tran} , and M_{Dura}). The remaining retrieval path, delayed retrieval from *durable* storage, is redundant for WLM tests with a single delayed free recall task, as it is the only way to recall on the delayed free recall task and is expressed in M_{DFR1} .

Evaluating ADNI ADAS-Cog WLM Test with qCPs

Relation to Disease Progression (Clinical Diagnosis)

The new qCPs were generated for each of 10,868 ADAS-Cog WLM tests from 2,366 participants enrolled in the ADNI, and the means of these qCPs were compared against clinical diagnoses (CN: cognitively normal; SCI: subjective cognitive impairment; MCI: mild cognitive impairment; AD: Alzheimer's disease). After accounting for varying degrees of item effects in the shuffled-order test, the results show the expected decline from CN, SCI, MCI to AD across participants assessments for many of the qCPs (**Figure 4**).



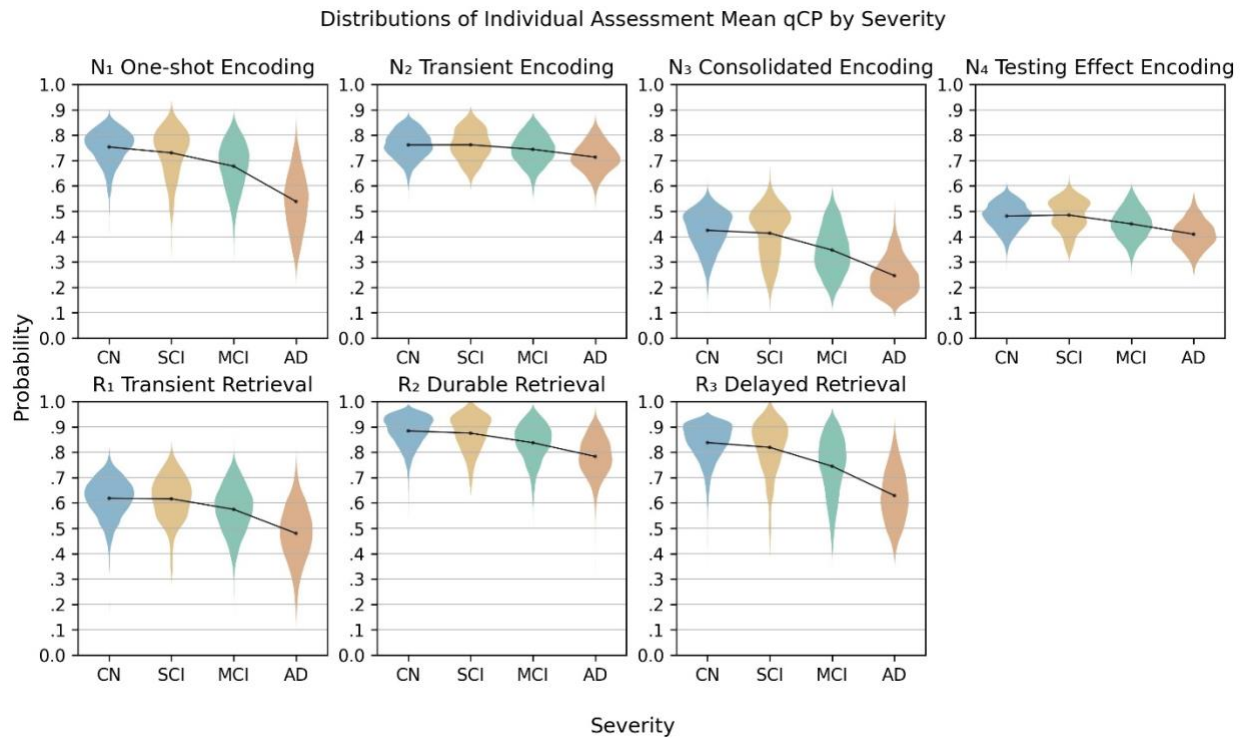


Figure 4. Distributions of the mean qCPs for assessments in the ADNI dataset, by Alzheimer's disease severity stage.

M Parameter Relation to Disease Progression (Clinical Diagnosis)

M parameters were likewise generated for each of the 10,868 ADAS-Cog WLM tests from 2,366 participants enrolled in the ADNI, and the means of these behavioral measures were compared against clinical diagnoses (CN, SCI, MCI, and AD). The adjusted parameters show the expected change across disease progression across participant assessments (**Figure 5**). Notably, M_{Tran} shows a small increase with increasing impairment; this characterizes the manner in which more impaired participants are recalling those few items that they do recall from *transient* storage, when their ability to recall from *durable* storage declines.

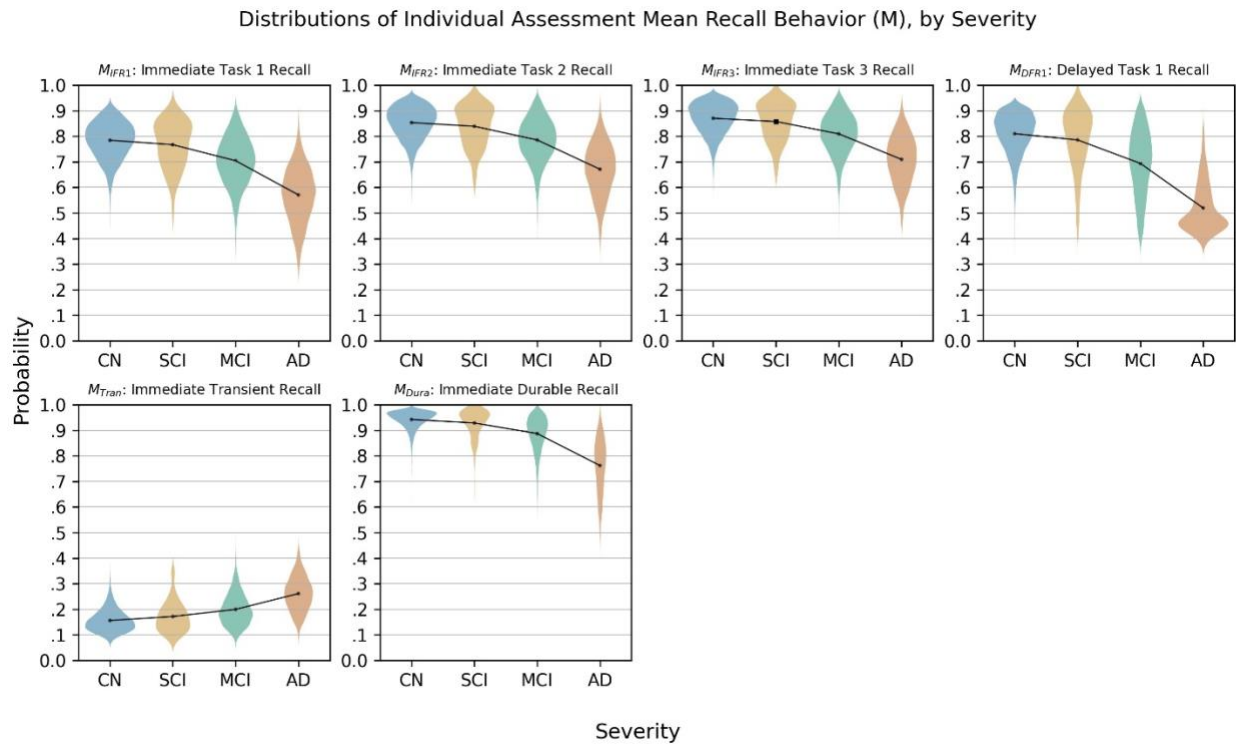


Figure 5. Distributions of the mean M 's for assessments in the ADNI dataset, by Alzheimer's disease severity stage.

Classification of Mild Cognitive Impairment due to AD with M Parameters

A Bayesian logistic regression model was used to assess the capability of the parameters generated with the HBCP-1 model to classify individuals as having MCI due to AD. From the ADNI dataset, participant assessments were classified as CN if CDR-GS = 0 and CSF amyloid, p-tau, and t-tau are negative (ATN-), according to ADNI criteria ($n = 726$); alternatively, participant assessments were classified as MCI if CDR-GS = 0.5 and ATN+ ($n = 1,115$). These groups were evenly split into training and testing subgroups. The Bayesian logistic regression model was run on the training subgroup of participants, and coefficients for age, sex, education, M_{IFR3} , and M_{DFR1} were obtained. These coefficients were applied to participants in the testing subgroup, and an ROC analysis was performed on the classification ($AUC = .91$) (**Figure 6**). At the cutoff $p = .76$ (Youden's $j = .69$), accuracy = .83, sensitivity = .76, and specificity = .93.

Bayesian Logistic Regression Classification of MCI from CN Participants with Demographics, M_{IFR3} , and M_{DFR1}

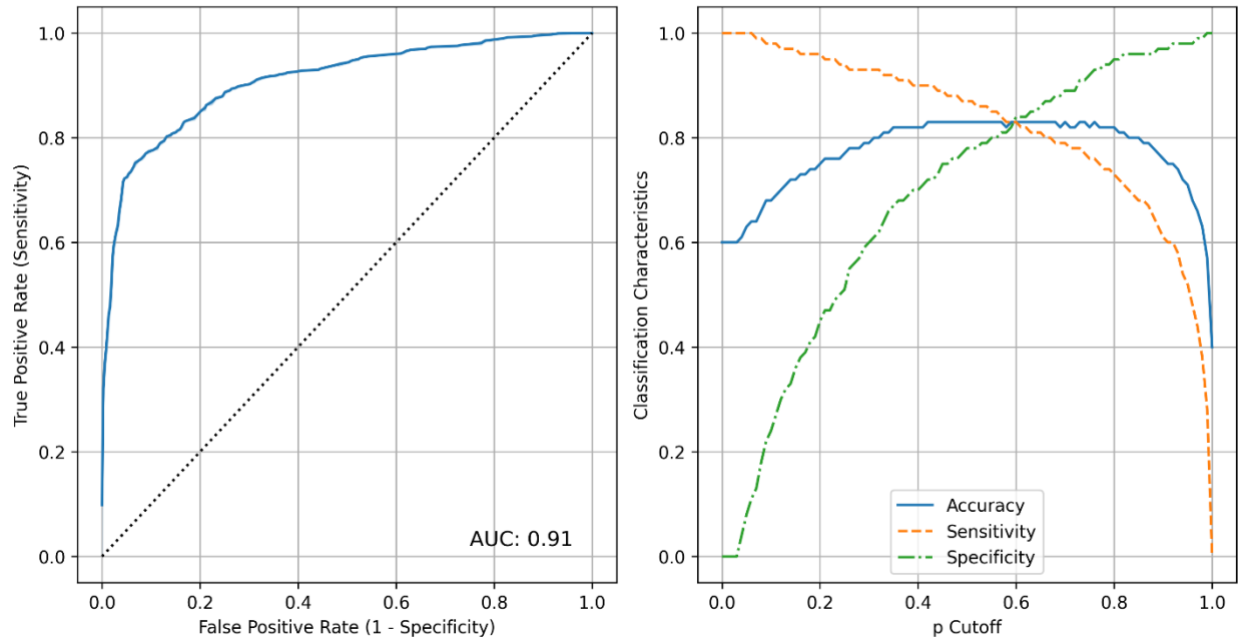


Figure 6. Classification statistics for the Bayesian logistic regression model, classifying MCI from CN participants with demographics, M_{IFR3} , and M_{DFR1} .

Dataset Information

This methods document applies to the following dataset(s) available from the ADNI repository:

Dataset Name	Date Submitted	Note
ADNI_EMBICDCB.csv	July 27, 2022	Previous data delivery
ADAS_EMBIC_qCP.csv	November 27, 2023	Current data delivery

The following parameters are provided in this dataset, with each including a field for the mean and a field for the standard deviation of the parameter posterior distribution of the individual assessment value, characterizing latent ability, adjusted for the impact of test features:

qCPs	Correlate	Description
N_1	Encoding	Latent probability of encoding into the DURABLY LEARNED State
N_2	Encoding	Latent probability of encoding into the TRANSIENTLY LEARNED State
N_3	Encoding	Latent probability of encoding into the DURABLY LEARNED State, following previous TRANSIENT LEARNING (N_2)



N_4	Encoding	Latent probability of encoding into the DURABLY LEARNED State, due to successful retrieval (R_1) from the TRANSIENTLY LEARNED State
R_1	Retrieval	Latent probability of retrieving from the TRANSIENTLY LEARNED State
R_2	Retrieval	Latent probability of retrieving from the DURABLY LEARNED State
R_3	Retrieval	Latent probability of retrieving from the DURABLY LEARNED State after a 5-minute delay with distraction
M_{IFR1}	Recall	Latent probability of recall of an episodic memory on immediate recall task 1
M_{IFR2}	Recall	Latent probability of recall of an episodic memory on immediate recall task 2
M_{IFR3}	Recall	Latent probability of recall of an episodic memory on immediate recall task 3
M_{DFR1}	Recall	Latent probability of recall of a durably stored episodic memory on delayed recall task 1
M_{Tran}	Recall	Latent probability of immediate recall of a non-durably stored episodic memory on any task
M_{Dura}	Recall	Latent probability of immediate recall of a durably stored episodic memory on any task

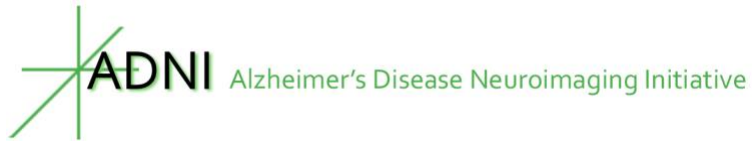
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